

Chapter 2

REVIEW OF RELATED LITERATURE AND RELATED STUDIES

This chapter reviews the literature and studies the researcher examined, providing essential knowledge, concepts, and ideas that support and clarify the direction of the current study.

Related Literature

Presence of Fare

Peters and Bhusal (2020) explained that paratransit services — a category that includes tricycles — are best understood as flexible, demand-driven forms of transport that typically operate without the fixed schedules, stops, and standardized pricing found in formal public transit systems. They noted that this flexibility, while beneficial for accessibility in underserved areas, often comes at the cost of consistent, transparent pricing structures that passengers can rely on. According to their analysis, most city authorities attempt to formalize paratransit by introducing regulatory tools such as fare structures, but enforcement capacity in many regions remains limited. This suggests that the presence of a formal pricing mechanism, such as a displayed fare matrix, becomes especially important precisely because paratransit's flexible nature makes fare abuse easier without it.

Dumedah, Abass, Gyasi, Forkuor, and Novignon (2023) examined paratransit service delivery in the Oforikrom Municipality of Ghana and found that fares for paratransit services were often inconsistent, varying by route, distance, travel time, and even passenger demographics, without being deliberately or transparently communicated to riders. The researchers explained that these disparities stemmed less from ill intent and more from a lack of clearly defined and consistently applied fare structures within the sector. They recommended that transport authorities establish a clear fare structure, apply it uniformly, and communicate the basis for any pricing differences in order to ensure fare

transparency for commuters. This directly supports the idea that making fare information visible and standardized — such as through a displayed matrix — is a practical solution to reducing pricing inconsistency in paratransit-type systems like tricycles.

Boquet and Boquet (2017) provided an overview of the Philippine transportation system, describing how local mobility across the archipelago relies heavily on distinctly Philippine vehicles such as the jeepney, tricycle ("trisikel"), and pedicab. They explained that unlike national transport policy, the regulation of these vehicles — including their fares and routes — has largely been devolved to local government units, resulting in fare-setting and enforcement practices that vary significantly from one municipality to another. This decentralized structure, they noted, means that the responsibility for ensuring fare transparency, such as through the mandatory posting of fare matrices, falls squarely on individual LGUs rather than a single national authority. This literature establishes the broader institutional context explaining why fare matrix display compliance can differ so widely between municipalities in the Philippines, including Sta. Cruz, Laguna.

Matrix Displays

Tufte, E. R. (2020). *Seeing With Fresh Eyes: Meaning, Space, Data, Truth*. Graphics Press. discussed matrix displays as a powerful method for organizing and presenting complex information. He explained that arranging data in rows and columns allows people to compare multiple variables at once and identify patterns easily. This suggests that matrix displays are effective tools for making large amounts of data more understandable Edward Tufte's 6 Data.

Ware, C. (2021). *Visual Thinking for Information Design* (2nd ed.). emphasized the importance of visual design in information processing. He stated that the human brain interprets information presented in grid and matrix formats faster than in linear text. This means that matrix displays can improve comprehension and decision-making when dealing with technical or research data.

O'Hagan, A. 2019. Expert Knowledge Elicitation: Subjective but Scientific. *American Statistician*, 73(S1), 69-81. - This is the 2019-2023 modern version of Meyer & Booker. highlighted decision matrices as essential tools in engineering and management. They explained that matrix displays help decision-makers systematically evaluate alternatives based on different criteria. Therefore, using matrices leads to more objective and structured choices.

Regulatory Monitoring

Rodrigue et al. (2025) examined the relationship between public transportation fare characteristics and fare equity through a scoping review of empirical studies. Their review found that different fare characteristics may produce different effects on transportation equity. The researchers emphasized that the effects of fare policies depend on factors such as passenger characteristics, transportation systems, and the manner in which fare policies are implemented.

Bruno et al. (2026) reviewed 58 studies concerning equity in public transportation fare policies. Their review examined different fare policies and found that their effects on transportation equity depend on factors such as the characteristics of passengers, the transportation system, and the implementation of the policy. The authors emphasized the importance of considering accessibility

and the needs of different passenger groups when developing transportation fare policies.

Ong et al. (2023) examined the service quality and passenger satisfaction of tricycle transportation in Philippine urban areas. Their study emphasized that the quality of tricycle services is influenced by several factors, including responsiveness, reliability, and the manner in which services are provided to passengers. Although the study primarily focused on service quality, its findings demonstrate the importance of effective regulation in maintaining acceptable standards in local public transportation.

Fairness of Fares

The concept of fare fairness is most commonly examined through equity. Rubensson, Susilo, and Cats (2020) developed an empirical framework assessing the geographical and distributional fairness of flat, zone-based, and distance-based fare schemes in Stockholm. They found that high-income travelers benefited least from flat fares, while distance-based schemes, though horizontally equitable, were more regressive from a vertical equity standpoint — a useful foundation for treating fairness as measurable rather than purely subjective.

Chen, Pougala, and Stathopoulos (2026) modeled perceived fairness among Chicago Transit Authority riders using an Error Component Logit approach on survey data. Fairness perceptions were significantly linked to gender, race, income, and familiarity with fare policy — relevant if your DV is a subjective, survey-measured construct.

Narag, Abeya, Pascua, and Tuppil (2025) examined informal fare bargaining practices between tricycle drivers and commuters in Tuguegarao City. They found

that a favorable or negotiated price is not automatically perceived as the fairest one, suggesting fairness judgments are distinct from simple price satisfaction and are instead rooted in a sense of equitable treatment — a finding that grounds the abstract equity and perception concepts above in a local, everyday commuting context.

Satisfaction with Regulatory Response

Mylonakou, Chassiakos, Karatzas, and Liappi (2023) developed a system dynamics model to evaluate how the transport sector, and by extension the mobility policies and regulatory actions applied to it, affects overall citizen satisfaction in Patras City, Greece. The model simulated citizen satisfaction outcomes across several regulatory action scenarios, including improved accessibility for alternative transport modes, enhanced travel safety measures, reduced emissions, and greater efficiency in the movement of people and goods. The study found that local decision-makers need to take further concrete measures to meaningfully increase citizen satisfaction, and proposed the model as a decision-support tool that can simulate how new policies or regulatory interventions would shift satisfaction levels before they are implemented. This study is relevant to the present research because it demonstrates a method for directly linking specific regulatory/policy actions to satisfaction outcomes, rather than treating satisfaction as a static endpoint disconnected from the regulatory decisions that produced it.

Sun, He, and Kejriwal (2026) analyzed over 1.27 million consumer financial complaints filed with the U.S. Consumer Financial Protection Bureau (CFPB) between 2014 and 2022 to examine whether the *timeliness* of a regulated company's response — a core measure of regulatory-system responsiveness —

differed systematically across demographic groups. The study found that while overall response timeliness improved for the general population over the study period, service members consistently experienced lower probabilities of a timely response, and older adults shifted from a slight advantage in 2014 to a measurable disadvantage by 2022, with disparities persisting even after accounting for neighborhood-level economic disadvantage. This study is highly relevant to "satisfaction with regulatory response" as a dependent variable because it demonstrates, at a large scale, that regulatory responsiveness is not uniformly experienced across a population — meaning satisfaction with a regulator's response may itself vary systematically by who is filing the complaint, not just by the substance of the complaint itself.

Canlas, Dayo, Mallari, Pangilinan, Salazar, Tolentino, Tanig Jr., and Lim (2023) conducted a descriptive-correlational study assessing the relationship between the quality of the public transportation system (jeepneys) and the effectiveness of transport policies in the City of San Fernando, Pampanga. Using survey and qualitative data gathered from commuters, jeepney drivers, and traffic enforcers, the researchers found that commuters were only "somehow satisfied" with both the overall service quality and the effectiveness of transport policy implementation, with vehicle condition and environmental impact receiving notably poor ratings despite jeepney drivers and traffic enforcers rating the same policies more favorably. The study further established a positive monotonic relationship between transportation system quality and the perceived effectiveness of transport policies, and proposed a transport management strategy based on these findings. This study is directly relevant to the present research as it demonstrates, within a Philippine local government setting, that satisfaction with regulatory/policy

response is shaped by a visible gap between how regulators and enforcers perceive their own effectiveness and how the commuting public actually experiences it.

Related Studies

Presence of Fare

Caoleng (2024) conducted a phenomenological qualitative study exploring the experiences of commuters with tricycle fare pricing in the Municipality of Concepcion, gathering narrative accounts to understand how passengers perceive and are affected by fare practices. The study found that tricycle fare pricing in the area exhibited notable irregularities, with fares often exceeding previous rates and varying inconsistently between trips, leading to expressed dissatisfaction among commuters regarding both fairness and affordability. Based on these experiences, the researcher concluded that implementing and consistently displaying a standardized fare matrix was necessary to protect the interests of both commuters and drivers. This finding directly supports the idea that the visible presence of an official fare matrix serves as a practical solution to the fare inconsistencies experienced by passengers in similarly under-regulated municipalities.

Caoleng (2024) conducted a phenomenological qualitative study exploring commuters' experiences with tricycle fare pricing in the Municipality of Concepcion, finding notable fare irregularities and passenger dissatisfaction, and concluded that a standardized, displayed fare matrix was necessary to protect both commuters and drivers.

Aydinan (2020) conducted a descriptive study on tricycle drivers' knowledge and compliance with traffic rules and regulations in Cabanatuan City, finding

moderate to high regulatory knowledge and high compliance when local enforcement and awareness efforts were consistently applied.

Matrix Display

Kim, J. et al. (2019) studied the use of LED matrix displays in classroom instruction. They found that students exposed to lessons using interactive matrix displays showed higher engagement and better information retention than those using traditional methods. This suggests that matrix displays can enhance learning outcomes in education.

Garcia, R.L. & Dela Cruz, M. (2020) conducted a study on the use of research matrix charts among undergraduate students in the Philippines. They discovered that students who organized their RRL and RRS using a matrix format were able to synthesize literature more effectively and identify research gaps. This shows that matrix displays support academic research and writing.

Chen, L. & Wang, Y. (2022) investigated the effectiveness of digital matrix displays in public communication. They found that dynamic matrix LED screens captured more audience attention and improved message recall compared to static posters. This implies that matrix displays are valuable for both educational and commercial communication. Design and Characterization of Active Matrix LED Microdisplays With Embedded Visible Light Community.

Regulatory Monitoring

Rodrigue et al. (2025) conducted a scoping review of studies examining the relationship between public transportation fare characteristics and fare equity. The researchers screened 511 studies and retained 24 quantitative empirical studies for analysis. Their findings showed that the effects of fare structures vary

depending on the characteristics of the fare system and the context in which it is implemented.

Caoleng (2024) conducted a phenomenological study entitled Tricycle Fare Price in the Municipality of Concepcion: A Commuters' Perspective. The study focused on the experiences of commuters regarding tricycle fare prices and the problems they encounter in relation to fare collection.

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Fairness of Fares

Shuman, Guo, and Caros (2024) introduced a journey-based equity analysis methodology to measure differences in transit convenience between income groups, combining on-board ridership survey data with passenger origin-destination data at the census-block level. Applying this method to the Massachusetts Bay Transportation Authority (MBTA) bus network in Greater Boston, they found a significant disparity in service convenience — measured through number of transfers per unit distance, transfer wait time, and travel time per unit distance — between low-income and high-income riders. Although this study centers on service convenience rather than the fare amount itself, it is relevant to fairness of fares research because it demonstrates that the *value received* for a fare (in terms of trip convenience) is not equally distributed across

income groups, adding a service-quality dimension to how "fairness" of a transit fare can be understood beyond price alone.

Caoleng (2024) examined tricycle fare pricing in the Municipality of Concepcion using a phenomenological qualitative approach centered on commuters' own narrative accounts of their fare experiences. The study found that fare pricing exhibited noticeable irregularities, with commuters reporting fares that were often higher than in previous years and inconsistent from one trip to the next, leading to dissatisfaction and concerns over affordability. Based on these findings, Caoleng recommended the implementation of a standardized fare matrix to protect both commuters and drivers from the disputes caused by inconsistent pricing. This study reinforces, in a separate Philippine locale, the finding also observed by Narag et al. (2025) in Tuguegarao City: that fairness of fares is judged not simply by price level but by the *consistency and predictability* of how that price is applied.

Gatarin (2024) examined the Philippine government's Public Utility Vehicle Modernization Program (PUVMP), which aims to replace aging jeepneys — long regarded as the "king of the road" and a cultural icon — with modern, technologically upgraded vehicles. The study argued that the fare increases required to amortize the cost of new jeepney units raise pressing fairness concerns for both drivers, who face new debt obligations under the program, and passengers, who must absorb higher costs, situating the issue within a broader discourse of labor and mobility justice rather than treating it as a simple pricing adjustment. Gatarin further critiqued the largely top-down, system-wide nature of the reform, cautioning that just transitions require incremental change, inclusive dialogue with informal transport workers, and continued industry support rather than an abrupt phase-out. This study is relevant to fairness of fares research

because it demonstrates that fare fairness in the Philippine context cannot be separated from the *policy environment* driving fare changes — a fare increase imposed by a modernization mandate carries different fairness implications than one arising from market forces alone.

Satisfaction with Regulatory Response

This performance audit examined how well the European Commission and national enforcement bodies protected passengers' rights under EU Regulation 261/2004 during COVID-19. It found many passengers were not fully informed of their rights, were often pressured into vouchers instead of legally-owed cash refunds, and that enforcement varied inconsistently across EU member states despite billions in state aid to airlines. As an independent audit of real-world regulatory enforcement during a crisis, it offers a rigorous benchmark for evaluating regulatory response in passenger protection.

This descriptive thesis, conducted from January 2018 to April 2019 at Iloilo International Airport, compared how ten Civil Aeronautics Board (CAB) personnel, ten airline employees, and ten air passengers each perceived the sufficiency and effectiveness of the Air Passenger Bill of Rights (APBR). The three-way comparison is directly relevant to satisfaction with regulatory response because it captures whether the CAB's enforcement is seen as adequate by the passengers it is meant to protect, not just by the regulator or the airlines.

This quantitative descriptive-correlational study surveyed 385 passengers across five CAAP-managed airports (Davao, Iloilo, Laguindingan, Bohol-Panglao, Bacolod-Silay) to determine how security measures, environmental sustainability, and service quality shape passenger satisfaction. Since security and service

standards at these airports are mandated and monitored by the Civil Aviation Authority of the Philippines (CAAP), the findings show how government-enforced passenger-protection standards translate into actual passenger satisfaction.